

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	22 July 2026
Team ID	
Project Name	Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau
Maximum Marks	4 Marks

Technical Architecture

Cosmetic Insights — end-to-end flow from raw data to published dashboard:

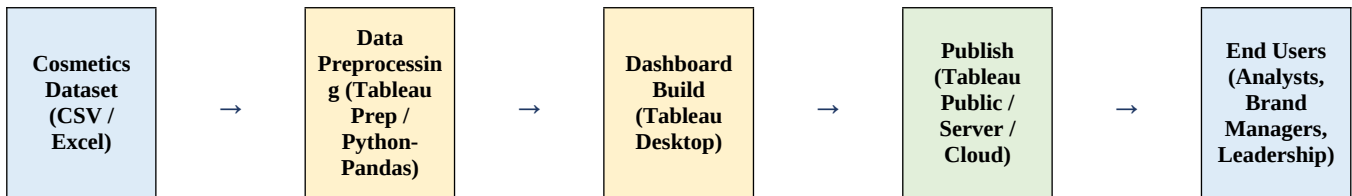


Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the dashboard	Tableau Desktop, Tableau Public / Server / Cloud (web view)
2.	Data Source	Raw cosmetics dataset: brand, category, price, rank, skin type	CSV / Excel file
3.	Data Preprocessing	Cleaning, standardizing, and structuring the dataset	Tableau Prep, Python (Pandas), Excel
4.	Application Logic-1	Calculated fields, parameters, and LOD expressions for analysis	Tableau Calculated Fields, LOD Expressions
5.	Application Logic-2	Filtering, drill-down, and interactivity logic	Tableau Filters, Actions, Parameters
6.	Database	Storage/config of the cleaned working dataset	Local extract file / MySQL (optional)
7.	File Storage	Storage for dataset and workbook files	Local filesystem, Google Drive
8.	Hosting / Publishing	Where the finished dashboard is published and shared	Tableau Public, Tableau Server, Tableau Cloud
9.	External API	Not required for the current scope	N/A
10.	Machine Learning Model	Not required — project is descriptive/exploratory analytics	N/A
11.	Infrastructure (Server / Cloud)	Local Server: Tableau Desktop on team laptops. Cloud: Tableau Cloud for publishing and sharing	Local (Windows/macOS) + Tableau Cloud

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Optional preprocessing scripts before loading into Tableau	Python, Pandas
2.	Security Implementations	Workbook and data-source permissions restrict who can view or edit pricing data	Tableau Server/Cloud role-based permissions
3.	Scalable Architecture	Single-tier BI dashboard; scales by extending the data source or connecting to a live database	Tableau live/extract connections
4.	Availability	Dashboard published to Tableau Public/Cloud for anytime, anywhere access	Tableau Cloud
5.	Performance	Tableau extracts and optimized calculated fields keep filtering and load times fast	Tableau Extracts, Data Source Filters

References

<https://c4model.com/>

<https://www.tableau.com/solutions/architecture>